

YouTube Kids Content Strategy: Predicting High-Performing Cocomelon Videos

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Introduction

Studios with dedicated children's programming are increasingly competing with YouTube-native channels. This project uses an 11-year catalog from Cocomelon to ask a practical question: what content, timing, and format decisions most predict whether a video performs above the channel's own median? The goal is a replicable framework a content programming team at a studio like Paramount's Nick Jr. or Netflix Kids could apply to its own YouTube strategy. Part B (unsupervised learning) ran first, identifying natural content groupings via sentence embeddings, UMAP, KMeans, and BERTopic. Those topic labels fed directly into Part A (supervised learning), where we trained and compared Logistic Regression, XGBoost, and KNN.

Our main findings show content type is the single most important predictor, with a 0.046 F1 drop when topic features are removed from XGBoost; video duration is second, with 5-10 minute videos performing best; XGBoost achieves the best overall F1 (0.708 CV) while KNN achieves the highest recall (0.83 test); and Cocomelon's catalog organizes into six content pillars, with Vehicles as the strongest consistent top performer across all three models. This project extends our Milestone I work (Solieman and Crossett, 2024), which analyzed five YouTube kids channels using EDA, regression, and decision trees. All supervised and unsupervised ML in this report is new; only data preprocessing and the average daily views metric are reused.

Related Work

The closest existing study is Saurabh and Gautam (2018), which used a decision tree classifier to identify which upload timing and video age features most predict viewership on a single YouTube educational channel. Our project improves on this in three ways: we compare multiple model families rather than a single classifier, we derive content topic features from unsupervised clustering rather than relying on metadata tags alone, and our analysis is designed with a business strategy audience in mind rather than a pure accuracy benchmark.

Alammar and Grootendorst (2024) provided the conceptual foundation for our BERTopic pipeline, specifically the chapters covering sentence embeddings, UMAP, and class-based TF-IDF topic representations. Grootendorst (2022) is also the original creator of the BERTopic library.

Covington, Adams, and Sargin (2016) described YouTube's deep neural network recommendation system, establishing that watch time and engagement signals are central to algorithmic placement. Our project addresses the upstream question of which content formats tend to generate those signals in the first place.

Data Source

We obtained our dataset through the Apify platform, which provides structured access to YouTube's Data API v3. The dataset covers all Cocomelon videos from channel inception through May 16, 2026, including video-level metrics such as view counts, upload timestamps, durations, titles, and description text.

Channel: Cocomelon (UCbCmjCuTUZos6Inko4u57UQ)

Timeframe: February 2015 through May 2026

Total videos: 1,216 raw; 1,162 after removing 54 records with null upload dates

Format: CSV; camelCase column names converted to snake_case for analysis

Key fields include: title, date, duration, viewCount, likes, text (video description containing lyrics or episode narrative, present for all 1,162 remaining records), and extractedAt.

The average daily views metric (total views divided by days since upload) was computed as the primary performance measure. This controls for video age and enables fair comparison across an 11-year catalog, where older videos would otherwise appear artificially stronger on raw view counts alone.

Feature Engineering

All features were derived from the cleaned dataset. Four videos with durations under 60 seconds were excluded as YouTube Shorts, leaving 1,158 videos for modeling. The target variable is a binary label (is_high_performer_tag) set to 1 for videos at or above the channel median of 30,295 average daily views. The median threshold was chosen over a top-quartile cutoff because it produces balanced classes (579 high performers, 579 low performers) and better reflects the practical question a programming team faces: which decisions reliably clear the bar, not which decisions occasionally produce a viral hit. Five features were engineered: upload day of week (one-hot encoded, Mon-Sun); upload season derived from upload week number (Winter/Spring/Summer/Fall); duration bucket across six ranges (0-2 min through 45 min+); upload count per week as a numeric upload-frequency signal; and topic, a content pillar label derived from Part B clustering and applied via keyword-based mapping (quick_map). Compilations are only assigned when duration is 10 minutes or longer to prevent single songs from miscategorization. The resulting distribution: Educational_Learning (529), Compilations (306), Vehicles (120), Educational_Routines (116), Special_Events (79), Character_Stories (8). Full encoding details are in the notebook.

Part A: Supervised Learning

Methods

Three model families were trained and compared. Each was chosen to represent a meaningfully different underlying mechanism. Logistic Regression provides a linear probabilistic baseline with directly interpretable coefficients, making it the most actionable model for a content team. XGBoost builds an ensemble of decision trees that captures non-linear feature interactions (such as how duration and upload day might combine), suited to cases where simple linear separability is unlikely. KNN is a non-parametric, instance-based model that makes no distributional assumptions, providing a useful contrast to the other two.

All three models used the same preprocessing pipeline: OneHotEncoder for categorical features (upload day, season, duration bucket, topic) and StandardScaler for the single numeric feature (upload count per week). A 70/30 stratified train/test split was applied with random_state=101, and GridSearchCV with 5-fold cross-validation was used for hyperparameter tuning on all three models.

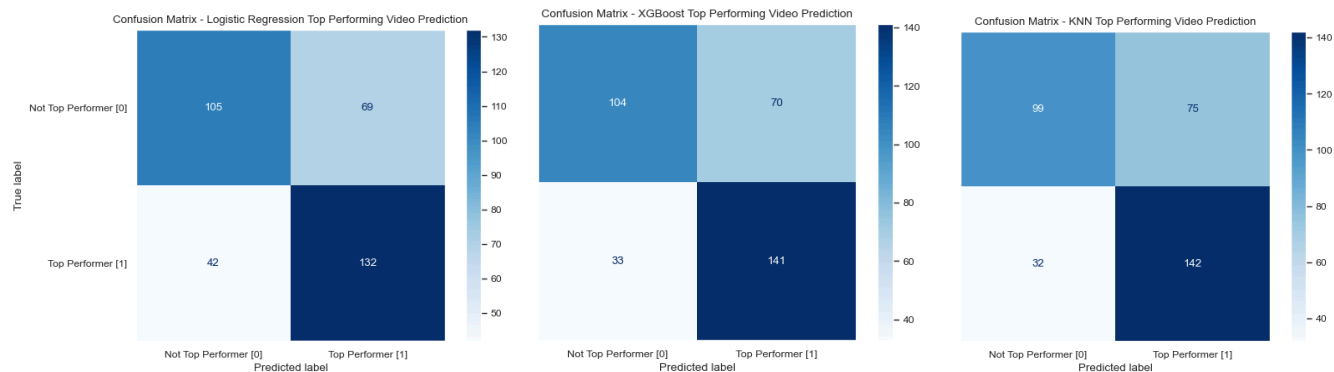
For Logistic Regression, C was searched over [0.01, 0.1, 1, 10, 100]; best value: C=100. For XGBoost, grid search over n_estimators, max_depth, learning_rate, and subsample; best configuration: learning_rate=0.1, max_depth=3, n_estimators=50, subsample=1.0. For KNN, n_neighbors was searched from 3 to 101 with uniform and distance weighting; best: n_neighbors=61, uniform weighting.

Overall Results

Model	CV Accuracy	CV F1	CV Recall	CV Precision	Test F1	Test Recall
Logistic Regression	0.663 +/- 0.029	0.690 +/- 0.030	0.751 +/- 0.042	0.638 +/- 0.023	0.70	0.76
XGBoost	0.679 +/- 0.043	0.708 +/- 0.040	0.778 +/- 0.051	0.650 +/- 0.037	0.73	0.81
KNN	0.648 +/- 0.034	0.692 +/- 0.032	0.790 +/- 0.046	0.615 +/- 0.026	0.72	0.83

Table 1. Model Comparison (5-Fold Cross-Validation on Training Set). XGBoost achieves the best overall F1 (0.708 CV, 0.73 test). KNN achieves the highest recall (0.790 CV, 0.83 test) at the cost of lower precision. Logistic Regression has the most balanced precision relative to the other two, making it the most conservative model in terms of false positives.

Confusion matrix figures for all three models: Figure 1 (LR), Figure 2 (XGBoost), Figure 3 (KNN).



Feature Importance and Ablation Analysis

Feature importance was evaluated three ways: Logistic Regression coefficients for directional signals, and an ablation study on XGBoost to quantify how much each feature group contributes, and a KNN ablation study to test whether the distance-based model relied on the same feature groups. The LR coefficients (Figure 4) show the strongest positive signals are 5-10 minute duration (+1.25) and the Vehicles topic (+1.00); the strongest negatives are 0-2 minute duration (-2.45), Character_Stories (-1.28), and Sunday uploads (-0.84). Upload frequency has a small negative signal (-0.23).

Feature weight table: Figure 4

	feature	weights
17	onehotencoder__duration_bucket_5-10min	1.248223
23	onehotencoder__topic_Vehicles	0.995897
14	onehotencoder__duration_bucket_2-5min	0.898287
6	onehotencoder__upload_day_of_week_Wednesday	0.651596
15	onehotencoder__duration_bucket_30-45min	0.601414
19	onehotencoder__topic_Compilations	0.477191
5	onehotencoder__upload_day_of_week_Tuesday	0.412685
0	onehotencoder__upload_day_of_week_Friday	0.385523
8	onehotencoder__upload_season_Spring (Apr-Jun)	0.364249
9	onehotencoder__upload_season_Summer (Jul-Sep)	0.332185
10	onehotencoder__upload_season_Winter (Jan-Mar)	0.240542
13	onehotencoder__duration_bucket_10-30min	0.216849
20	onehotencoder__topic_Educational_Learning	0.203417
21	onehotencoder__topic_Educational_Routines	0.050399
7	onehotencoder__upload_season_Fall (Oct-Dec)	0.039344
4	onehotencoder__upload_day_of_week_Thursday	-0.133900
1	onehotencoder__upload_day_of_week_Monday	-0.182592
24	standardscaler__upload_count_per_week	-0.233545
2	onehotencoder__upload_day_of_week_Saturday	-0.287723
22	onehotencoder__topic_Special_Events	-0.439920
16	onehotencoder__duration_bucket_45min+	-0.511721
3	onehotencoder__upload_day_of_week_Sunday	-0.837672
11	onehotencoder__upload_season_nan	-0.968402
18	onehotencoder__topic_Character_Stories	-1.279066
12	onehotencoder__duration_bucket_0-2min	-2.445133

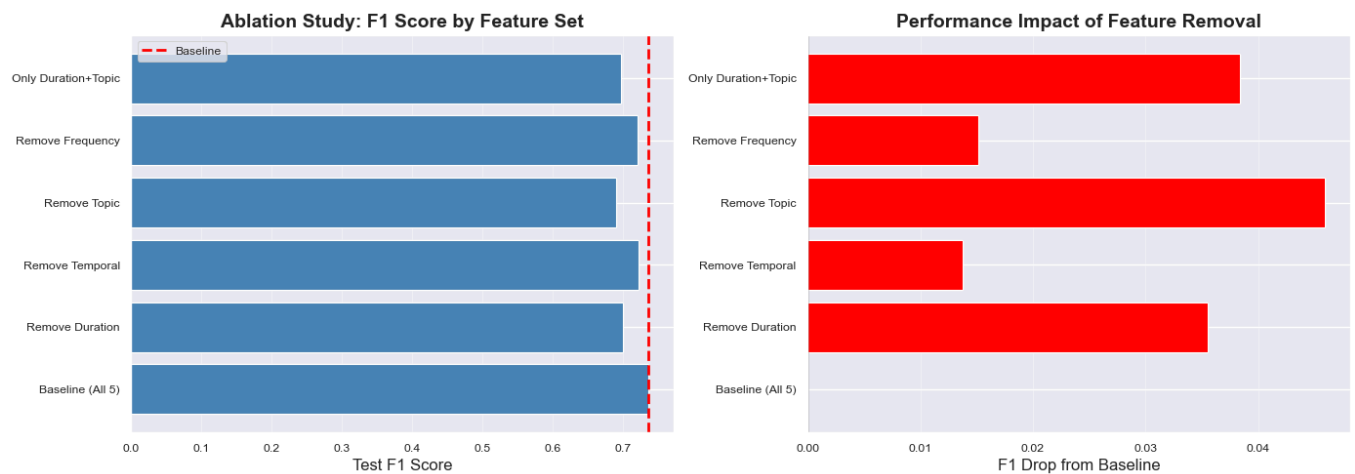
The ablation study systematically removed feature groups from XGBoost and measured the resulting F1 drop on the test set. The XGBoost results are shown in Table 2a.

Table 2a. XGBoost Ablation Study Results

Feature Set	Features	Test F1	F1 Drop
Baseline (All 5)	5	0.736	0.000
Remove Duration	4	0.700	0.036
Remove Temporal	3	0.722	0.014
Remove Topic	4	0.690	0.046
Remove Frequency	4	0.721	0.015
Only Duration + Topic	2	0.697	0.038

Topic is the most important feature group, causing the largest F1 drop (0.046) when removed. Duration is second (0.036 drop). Temporal features (upload day and season) and upload frequency each contribute only about 0.014-0.015. A simplified two-feature model using only Duration and Topic achieves an F1 of 0.697, which is 94.7% of baseline performance.

Ablation study bar charts - Figure 5



The same ablation setup was applied to KNN. Most feature-removal settings produced similar results (F1: 0.715-0.722), with the largest drop when duration was removed (F1: 0.697). This differs from XGBoost, where Topic drove the biggest drop, suggesting the tree-based model relies more on content category while KNN is more sensitive to video length.

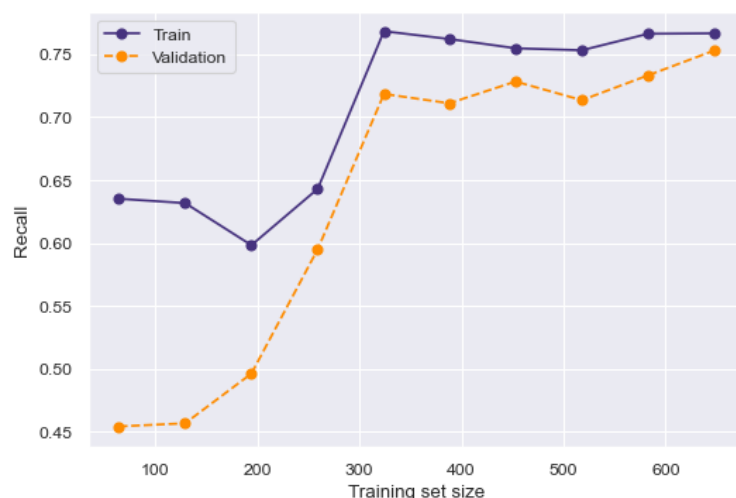
Table 2b. KNN Ablation Study Results

Feature Set	Features	Test F1	F1 Drop
Baseline (All 5)	5	0.722	0.000
Remove Duration	4	0.697	0.025
Remove Temporal	3	0.722	0.000
Remove Topic	4	0.718	0.003
Remove Frequency	4	0.715	0.006
Only Duration + Topic	2	0.717	0.005

Sensitivity Analysis

Logistic Regression was evaluated via a learning curve showing recall across training set sizes from roughly 80 to 810 examples. The validation recall stabilizes near the training recall and both curves converge by the full training set, with performance plateauing around 300 examples. This confirms the model is not overfitting and that collecting more data alone is unlikely to improve performance without improving feature quality. Separately, the C=100 result from the LR grid search (vs. C=10 and C=1) produced only marginal recall differences (<0.01), confirming the model is not sensitive to regularization strength in this range. XGBoost showed similar stability across the max_depth sweep (3-5), consistent with the grid search results.

Learning curve figure - Figure 6



Tradeoffs

Recall vs. precision: KNN achieves the highest recall (0.83) but the lowest precision (0.615). Logistic Regression offers a more balanced tradeoff (0.638 precision, 0.76 recall). A team with limited bandwidth should favor LR to reduce false alarms; a team doing broad ideation should favor KNN or XGBoost to reduce missed opportunities.

Model complexity vs. interpretability: LR provides directly readable coefficients (for

example, Vehicles topic = +1.00, 5-10 min duration = +1.25). XGBoost achieves the best F1 but requires SHAP to explain its decisions. The XGBoost ablation study shows the simplified two-feature model (Duration + Topic) gets within 5% of baseline performance while being far easier to explain to stakeholders.

Training data size vs. performance: The LR learning curve plateaus around 300 examples, suggesting diminishing returns from collecting more data without also improving feature quality, particularly topic classification, which drives the largest ablation impact.

Failure Analysis

Three specific failure categories were identified from XGBoost's test set misclassifications. These examples come from the test set; titles are drawn from the actual records.

Category 1 (False Positive, metadata-correct but underperformed): "Reading Song + More Nursery Rhymes and Kids Songs." This video was a 30-45 minute Compilation uploaded on a Friday in Spring, all features the model associates with success. It averaged only 26,204 views per day and was labeled low performer. The model correctly identified the structural signals but had no way to measure content quality, thumbnail appeal, or competition on the upload date. This is the most common failure mode: good metadata, unexceptional performance.

Category 2 (False Negative, surprise hit): "JJ's Show and Tell Day at School + More Nursery Rhymes." A 45-minute Compilation uploaded on Friday in Fall, predicted as a low performer but averaging 38,804 daily views. SHAP analysis showed the 45 min+ bucket has high variance in its contribution, meaning these very long videos can either succeed strongly or fail depending on context the model cannot see. This video likely benefited from strong algorithmic placement or a topical hook that drove discovery.

Category 3 (Edge case, broke all rules): "Christmas Songs Medley + More Nursery Rhymes." A 10-30 minute Special Events video uploaded on Sunday in Fall, with every feature the model associates with poor performance. It averaged 93,116 daily views. Holiday content with perennial search demand can overcome unfavorable upload timing, but the model has no visibility into seasonal search trends or evergreen catalog appeal.

Proposed improvements to address these failure categories: (1) add a thumbnail quality proxy using resolution metadata from the YouTube API; (2) incorporate a competition signal measuring how many

other kids channels uploaded on the same day; (3) add a seasonal search demand feature using Google Trends data for keywords like "christmas songs for kids."

Part B: Unsupervised Learning

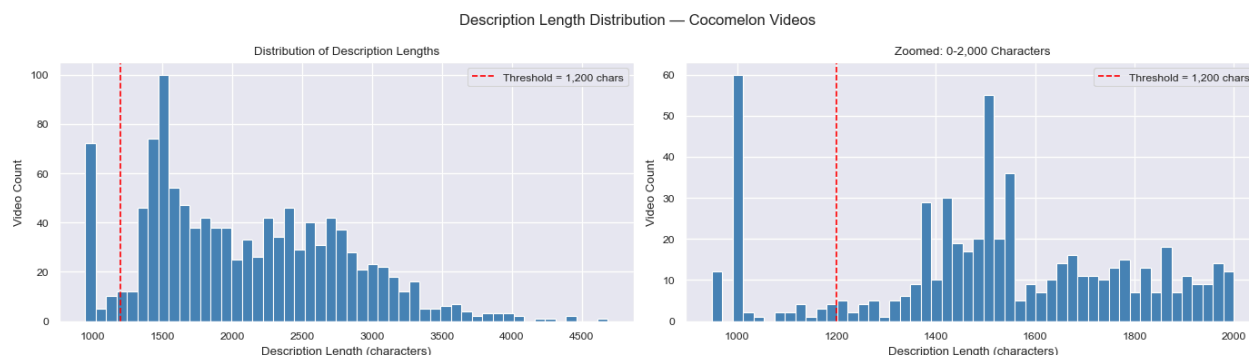
Methods

Part B ran before Part A and served two purposes: identifying natural content groupings in Cocomelon's catalog, and producing topic labels that became the most important feature in the supervised models.

The pipeline has four stages: text preprocessing, sentence embedding, dimensionality reduction, and clustering plus topic extraction.

Text preprocessing: Video descriptions vary widely in quality. About 92% of videos (1,067 of 1,158) have substantive descriptions containing full song lyrics or episode narratives. The remaining 91 videos have short boilerplate descriptions under 1,200 characters. A threshold of 1,200 characters was chosen to separate these two types after inspecting the distribution of description lengths. Videos above the threshold had their footer text stripped (streaming links, social handles, and copyright notice) and were embedded using title plus cleaned description. Videos below the threshold used the cleaned title only.

Description length distribution histogram - Figure 7



All 1,158 videos were embedded using sentence-transformers/all-mpnet-base-v2, producing 768-dimensional dense vectors. This was chosen over TF-IDF because TF-IDF treats "wheels on the bus" and "bus wheels" as semantically unrelated. UMAP was then applied twice: to 10 dimensions for clustering ($n_components=10$, $min_dist=0.0$, cosine metric) and to 2 dimensions for visualization only.

KMeans was run on the 10-dimensional embeddings with k values from 2 through 10. $k=4$ was selected because it produced reasonably sized clusters (103, 761, 152, and 142 videos); $k=5$ produced one cluster containing 49% of all videos and $k=6$ introduced a cluster of only 10. BERTopic was then run at $k=10$ using the same structure with c-TF-IDF keyword extraction added, giving each cluster a human-readable label based on statistically distinctive keywords. The silhouette score at $k=10$ (0.432) is higher than KMeans at $k=4$ (0.366), confirming the data supports finer granularity. The two methods are complementary: KMeans confirms natural groupings exist; BERTopic adds the keyword evidence needed to name them. The 10 topics were then manually collapsed to 6 content pillars for business use.

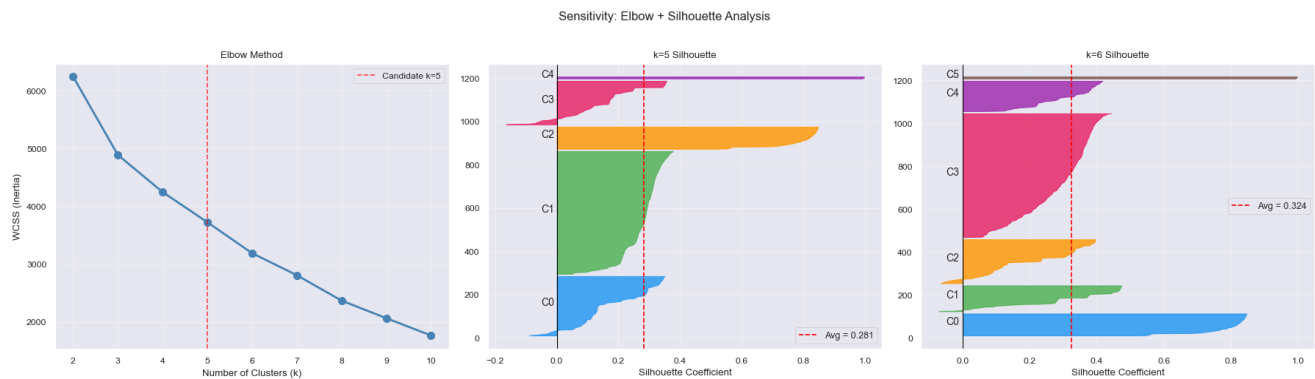
Unsupervised Evaluation

Table 3. Clustering Evaluation

Method	k	Silhouette Score
KMeans	4	0.366
BERTopic	10	0.432

Silhouette scores in the 0.3-0.5 range are typical for short-text clustering where semantic overlap between categories is expected. A bath song can also contain character dialogue; a holiday video can be educational. Perfect separation is not expected, and the scores we observe confirm that real groupings exist without overstating how clean those groupings are.

Elbow and silhouette plots - Figure 8

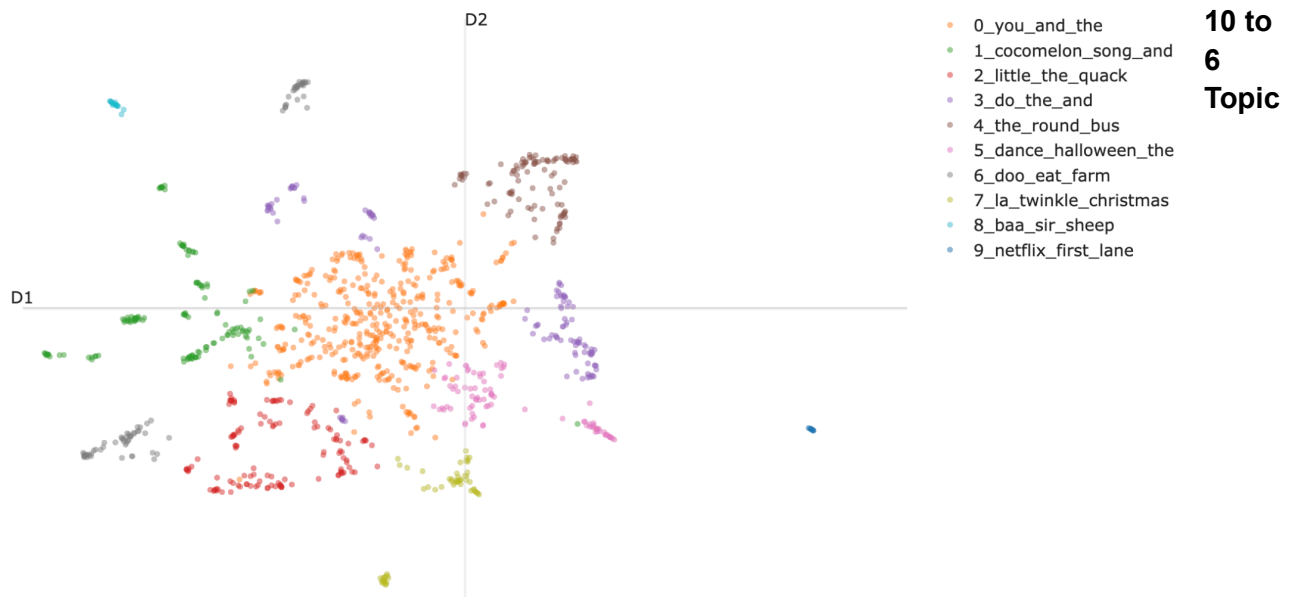


KMeans UMAP scatter plot - Figure 9



BERTopic semantic space scatter plot - Figure 10

BERTopic Topics in Semantic Space (k=10)



Collapse

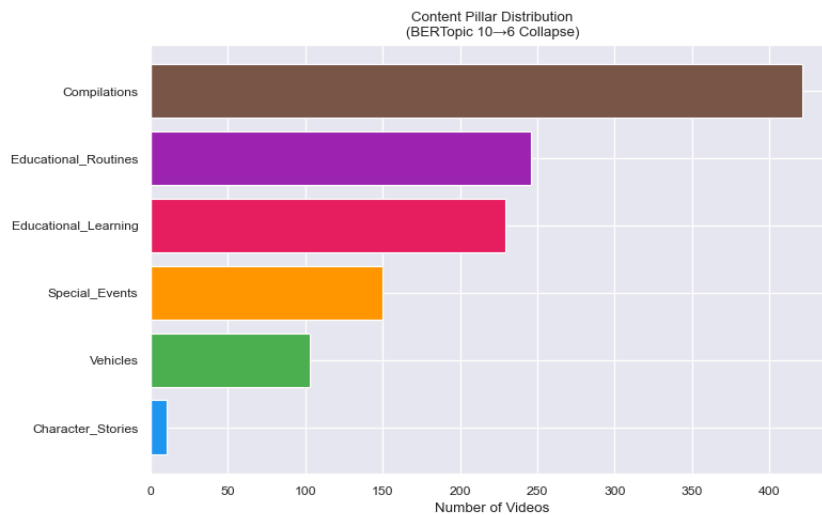
The 10 BERTopic topics were manually reviewed by inspecting the representative documents and keyword lists for each cluster. Topics were collapsed to 6 content pillars based on thematic similarity and strategic usefulness. The mapping is shown in Table 4.

Table 4. BERTopic 10 to 6 Collapse

BERTopic Topic	Count	Assigned Pillar
0: Compilations	426	Compilations
1: Holiday/Seasonal	133	Special_Events
2: Classic Nursery Rhymes	129	Educational_Learning
3: Water/Animal Rhymes	117	Educational_Routines
4: Wheels on the Bus	103	Vehicles
5: Dance/Activity	85	Educational_Routines
6: Farm/Food	78	Educational_Learning
7: Baa Baa/Animal	60	Educational_Routines
8: Twinkle/Lullabies	17	Educational_Routines
9: Netflix/CoComelon Lane	10	Character_Stories

Note: BERTopic's raw Compilations count (426) differs from the 306 used in Part A. The Part A `quick_map()` function applied a duration filter (minimum 10 minutes) to validate that Compilations are genuinely long-form content. Videos that matched no keyword pattern and were under 10 minutes were reassigned to Educational_Learning. This is intentional: Part B identifies natural semantic groupings, while Part A applies domain judgment to produce a cleaner, more defensible feature.

Content Pillar Distribution bar chart - Figure 11



Discussion

Part A

The most useful finding from Part A is not which model performed best but which features drove performance. Topic and duration account for the majority of predictive signal, while timing and upload frequency each add only marginal gains. This has a direct implication for a content team: invest in refining what you make and how long it is

before optimizing when you post.

The biggest surprise was the durability of the Vehicles finding. Despite Cocomelon releasing Wheels on the Bus variations repeatedly for over a decade, the format's LR coefficient (+1.00) is the second-strongest positive signal in the model, suggesting it taps something fundamental in preschool viewing behavior, possibly the combination of repetition, motion, and call-and-response structure. The main challenge was feature scope. With only five features, all derived from metadata rather than content, the models cannot see anything about what actually happens in a video: the animation quality, the hook in the first 10 seconds, the thumbnail. The three failure categories from the XGBoost analysis illustrate this ceiling clearly. The SHAP analysis added nuance here by showing that the 45 min+ duration bucket has high variance in its contribution, meaning it is not uniformly good or bad but rather context-dependent in ways the current feature set cannot fully capture. With more time, the most valuable extension would be integrating YouTube Thumbnail API metadata to proxy visual quality, and Google Trends data to capture seasonal search demand for specific keywords.

Part B

The most useful finding from Part B was that the catalog has a genuinely distinct semantic structure. Vehicles content emerged as the tightest cluster, confirmed by both methods, with consistent vocabulary across 11 years. Compilations (306 validated videos, 37.4-minute median) confirmed a real long-form publishing strategy rather than miscategorized singles. Educational_Routines is the highest-volume single-song category but shows near-neutral performance, functioning as catalog filler. Special_Events carries the highest underperformance risk due to seasonal replay limitations. Character_Stories' negative signal reflects a measurement mismatch: these Netflix-distributed episodes are benchmarked against a YouTube daily views metric they were never designed for. The biggest challenge was describing boilerplate. Cocomelon's descriptions contain a standard footer with streaming links, social handles, and copyright text that appears in nearly every video and adds no

content signal. The 1,200 character threshold cleanly separates videos where substantive text exists from those with only boilerplate, but the repeated footer still affects keyword quality in BERTopic's auto-generated labels. Stop words dominated several topic representations even after the footer was stripped. The actual content of each topic was clearer from the representative documents than from the keyword labels, which is why the 10 to 6 collapse was done manually with reference to both sources of evidence. With more time, one natural extension would be running BERTopic at a finer granularity ($k=15$ or $k=20$) to see whether sub-categories within Educational_Learning (for example, separating number/letter songs from food and farm songs) carry meaningfully different performance signals.

Ethical Considerations

Part A

The primary ethical risk in deploying a video performance classifier is that it reinforces existing patterns rather than enabling genuine creativity. If a content team uses model predictions as gates on which concepts get greenlit, they may systematically underinvest in formats the model has not seen succeed before, creating a feedback loop that shrinks the range of content children are exposed to. This is a real concern in any recommender or classifier that is applied to creative production decisions. There is also a fairness question around which baseline is used. The model is trained on Cocomelon's own performance distribution. Content from smaller channels, newer formats, or underrepresented cultural contexts would be measured against a benchmark derived from one of the most-viewed channels in YouTube history.

Part B

The unsupervised topic model raises a different concern: if the content pillars it identifies are used to shape a channel's publishing strategy over time, the model's own categories become self-reinforcing. A studio that bets heavily on Vehicles because the model flags it as a strong performer may crowd out Educational_Routines content even if that content serves important developmental functions for the children watching it. More broadly, the metrics we optimize for (average daily views) are correlated with platform recommendation algorithms, not directly with child learning outcomes or wellbeing. A video that scores high on this metric may do so because it is highly re-watchable or because its title matches search trends, neither of which necessarily means it is the most valuable content for a preschool audience.

Statement of Work

Michael Solieman: Data preparation, feature engineering, Logistic Regression (Part A Method 1), KMeans and BERTopic unsupervised analysis (Part B), topic-to-feature handoff, failure analysis, sensitivity analysis and report writing.

Sriram Penumatcha: XGBoost pipeline, cross-validation results, and feature importance analysis (Part A Method 2), SHAP analysis, ablation study, model comparison table and report writing.

Zhou Jiang: KNN pipeline, cross-validation results, and confusion matrix analysis (Part A Method 3), KNN ablation study, model comparison table, and report writing.

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